

ITHACA TOMPKINS, NY, USA

WMO: 725155

Lat: 42.483N

Lon: 76.467W

Elev: 335

StdP: 97.36

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 94761

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-15.2	-22.1	0.5	-17.0	-19.5	0.7	-14.4	12.1	-6.2	11.2	-4.8	3.6	80	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	31.3	22.4	29.7	21.6	28.2	20.8	23.8	29.2	22.9	27.8	22.0	26.3	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.5	26.4	21.3	16.6	25.6	20.6	15.9	24.9	72.7	29.1	69.3	27.7	66.0	26.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	8.0	DB	-22.7	34.3	2.8	1.4	-24.8	35.3	-26.4	36.1	-28.0	36.9	-30.0	37.9	(4)
(5)			WB	-22.8	25.4	2.8	1.0	-24.8	26.2	-26.4	26.7	-28.0	27.3	-30.0	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-4.5	-4.1	0.2	7.4	14.2	18.4	21.1	20.1	16.5	10.1	3.8	-1.1	(6)	
	DBStd	10.41	6.45	6.16	6.30	5.63	5.04	3.79	3.07	3.10	4.22	4.90	5.17	5.04	(7)	
	HDD10.0	1890	450	396	311	116	17	0	0	0	2	61	194	345	(8)	
	HDD18.3	3832	707	628	562	332	148	46	10	18	86	258	436	602	(9)	
	CDD10.0	1369	1	1	7	36	146	252	344	312	196	64	9	1	(10)	
	CDD18.3	271	0	0	0	3	20	48	96	72	29	3	0	0	(11)	
	CDH23.3	2371	0	0	1	39	217	434	847	565	247	21	0	0	(12)	
	CDH26.7	677	0	0	0	11	53	126	268	149	68	2	0	0	(13)	
(14) Wind	WSAvg	3.5	4.3	4.4	4.1	4.1	3.2	2.8	2.5	2.5	2.6	3.3	3.8	4.1	(14)	
(15) Precipitation	PrecAvg	916	52	49	62	78	84	97	88	87	96	80	80	62	(15)	
	PrecMax	1213	162	97	140	207	205	300	214	215	232	212	163	124	(16)	
	PrecMin	644	13	6	9	20	24	27	23	27	19	5	18	11	(17)	
	PrecStd	147	29	24	26	37	39	43	46	38	43	45	33	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	16.7	22.0	28.0	30.7	32.4	33.3	32.4	31.6	26.0	19.7	15.8	(19)	
		MCWB	11.6	11.9	13.7	16.2	20.6	23.1	23.6	23.2	22.2	19.7	13.8	13.6	(20)	
	2%	DB	10.9	11.1	16.4	23.5	28.4	30.1	31.3	30.2	28.6	22.7	16.5	11.3	(21)	
		MCWB	7.8	7.1	10.4	14.7	19.6	22.0	22.7	22.1	21.7	17.3	11.7	9.2	(22)	
	5%	DB	7.7	7.9	13.1	19.9	26.0	28.0	29.6	28.2	26.3	20.4	14.2	8.4	(23)	
		MCWB	5.6	5.4	9.1	12.5	18.5	20.8	21.8	21.0	20.3	16.7	10.1	6.5	(24)	
	10%	DB	4.5	4.4	10.0	17.0	23.4	26.0	27.9	26.7	23.8	17.9	12.0	6.2	(25)	
		MCWB	2.9	2.1	6.4	11.3	16.8	19.4	21.1	20.3	19.0	14.8	8.9	4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.0	14.9	18.0	22.6	24.0	25.1	24.6	23.9	20.9	14.7	13.4	(27)	
		MCDB	14.2	16.0	19.8	25.0	27.9	30.3	30.9	30.0	29.2	24.0	17.5	15.5	(28)	
	2%	WB	8.6	7.9	11.9	15.6	21.0	22.9	23.9	23.4	22.4	18.9	12.8	9.4	(29)	
		MCDB	9.8	10.4	15.5	21.4	27.2	28.7	29.5	27.9	27.1	21.2	15.8	11.0	(30)	
	5%	WB	5.5	5.4	9.4	13.8	19.4	21.7	22.9	22.3	21.1	17.2	10.9	6.9	(31)	
		MCDB	7.3	7.9	12.8	18.8	24.0	26.4	27.8	26.4	25.2	19.5	13.4	8.3	(32)	
	10%	WB	2.8	2.3	6.5	11.8	18.0	20.5	22.0	21.3	19.7	15.2	9.1	4.7	(33)	
		MCDB	4.1	4.4	9.8	16.1	22.1	24.5	26.0	25.1	22.6	17.8	11.6	6.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	9.8	11.5	12.1	11.4	11.8	11.4	11.4	9.6	8.6	6.9	(35)	
		MCDBR	10.0	12.2	13.8	16.6	14.7	14.0	13.6	13.1	13.4	12.6	12.3	9.6	(36)	
	5% WB	MCWBR	8.9	9.7	9.2	9.3	7.5	7.1	6.1	6.1	6.7	7.7	8.3	8.4	(37)	
		MCWBR	9.7	11.7	12.3	14.1	12.7	11.9	11.5	11.2	11.9	10.6	10.2	9.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.312	0.334	0.361	0.385	0.435	0.441	0.426	0.364	0.349	0.307	0.287	(40)	
		taud	2.372	2.325	2.378	2.364	2.335	2.214	2.210	2.240	2.398	2.442	2.514	2.468	(41)	
	Ebn at Noon		845	887	910	906	889	841	832	833	867	836	832	829	(42)	
		Edh at Noon	80	99	107	117	124	140	140	131	104	88	70	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.39	2.25	3.31	4.36	5.25	5.57	5.70	5.08	4.06	2.54	1.64	1.10	(44)	
	RadStd		0.13	0.33	0.30	0.51	0.52	0.52	0.44	0.29	0.41	0.28	0.21	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page